From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 198**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 14) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (15 to 20) contains 6 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

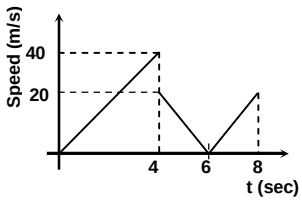
PART – I

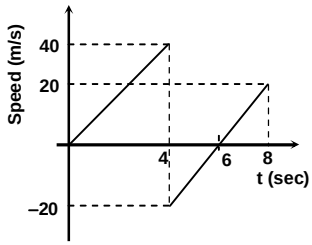
SECTION – A

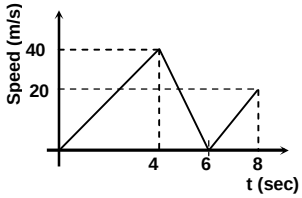
Single Correct Choice Type

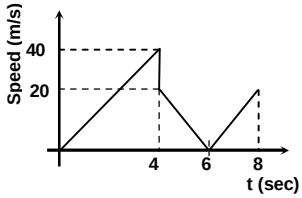
This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A ball is dropped from a height of 80 m on a floor. At each collision, the ball losses half of its speed. Then the graph of speed versus time is

(A)


(B)


(C)


(D)


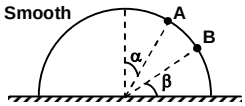
2. A projectile is required to hit a target whose co-ordinates relative to horizontal and vertical axes through the point of projection are (α, β). If the gun velocity is $\sqrt{2g\alpha}$. Then the condition when it is impossible to hit the target.

(A) $3\alpha > 4\beta$ (B) $4\beta > 3\alpha$

(C) $3\beta > 4\alpha$ (D) $4\alpha > 3\beta$

Space for Rough work

3. A particle moves from rest at A on the surface of a smooth circular cylinder of radius r as shown. At B its leaves the cylinder. Then the equation relating α and β is



(A) $3\sin\alpha = 2\cos\beta$ (B) $2\sin\alpha = 3\cos\beta$

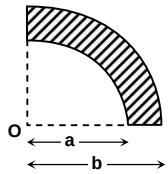
(C) $3\sin\beta = 2\cos\alpha$ (D) $2\sin\beta = 3\cos\alpha$

4. The gravitational field in a region is given by $E = \left(2\hat{i} + 3\hat{j}\right)$ N/kg. The equation of the line on which no work is done by the gravitational when the particle is moved

(A) $2y + 3x = 5$ (B) $5y + 2x = 3$

(C) $3y + 2x = 5$ (D) $3y + 5x = 2$

5. The distance of centre of mass from O of a uniform plate having quarter circular inner and outer boundaries of radii a and b from centre O.



(A) $\frac{4}{3\pi}\left(\frac{b^3 - a^3}{b^2 - a^2}\right)$ (B) $\frac{4\sqrt{2}}{3\pi}\left(\frac{b^3 + a^3}{b^2 - a^2}\right)$

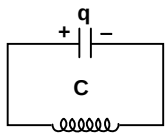
(C) $\frac{4}{\pi}\left(\frac{b^3 + a^3}{b^2 + a^2}\right)$ (D) $\frac{4\sqrt{2}}{3\pi}\left(\frac{b^3 - a^3}{b^2 - a^2}\right)$

6. Two concentric conducting spheres of radii R and 3R carrying charges Q and 2Q respectively. If the charge on inner sphere is doubled. The potential difference between inner and outer spheres will

(A) becomes two times (B) becomes four times

(C) be halved (D) remains same

7. In an L-C circuit shown in the figure, C = 1F, L = 4H. At time t = 0, charge in the capacitor is 4C and it is decreasing at a rate of $\sqrt{5}$ C/s. Choose the correct statements.



(A) maximum charge in the capacitor can be 6C

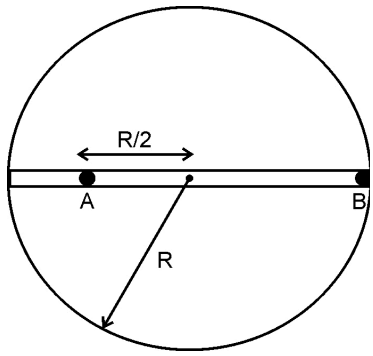
(B) maximum charge in the capacitor can be 8C

(C) charge in the capacitor will be maximum after time $2\sin^{-1}(2/3)$ sec

(D) None of these

Space for Rough work

8. A massive planet of radius R has a diametrical hole as shown in figure, such that hole does not effect the spherecity of the planet. The planet has uniform density. Two equal mass A and B [m << M] are simultaneously released, from positions shown. If the collisions are elastic then the total distance traveled by mass A after being released at the time of 6th collision is : [M→mass of planet, m→ mass of A and B neglect the gravitational interaction between mass A and mass B.]



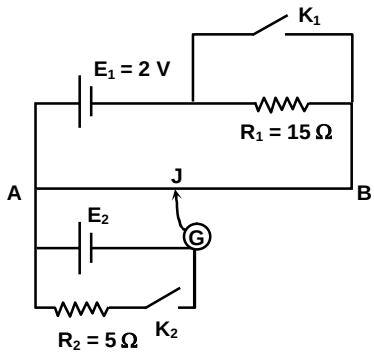
(A) 8.5 R (B) 5R
(C) 6.5 R (D) 9.5 R

Comprehension Type

This section contains 3 groups of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Questions 9 and 10

In the circuit shown AB is a 10 Ω uniform slide wire 50 cm long. E_1 is 2 V accumulator of negligible internal resistance R_1 and R_2 are 15 Ω and 5 Ω respectively. When k_1 and k_2 are both open, the galvanometer shows no deflection when AJ = 31.25. When k_1 and k_2 are both closed the balance length AJ = 5 cm



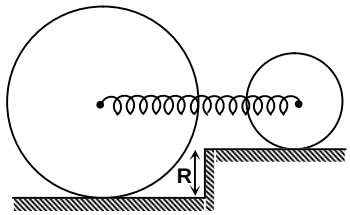
9. The emf of the cell E_2
(A) 0.5 V (B) 1 V
(C) 1.5 V (D) 2 V

10. The internal resistance of the cell E_2
(A) 7.5 Ω (B) 8 Ω
(C) 10 Ω (D) 2 Ω

Space for Rough work

Paragraph for Question Nos. 11 to 12

Two uniform circular disc of masses M and 4M and radii R and 2R respectively are connected by a massless spring of force constant k and placed in a vertical plane on consecutive steps of a stair as shown. Initially spring is compressed and system is released from rest. Assume that friction is enough to prevent slipping.



11. When spring is compressed by x, what is the friction force from stair on the mass 4M
(A) Kx/2 left
(B) kx/2 right,
(C) kx/3 left,
(D) kx/3 right

12. What is the angular frequency of oscillation of the system
(A) $\sqrt{\frac{5k}{6M}}$
(B) $\sqrt{\frac{5k}{4M}}$
(C) $\sqrt{\frac{k}{5M}}$
(D) none of these

Paragraph for Question Nos. 13 to 14

A ring of radius R is made of a thin wire of material of density ρ, having cross–section area a and Young's Modulus y. The ring rotates about an axis perpendicular to its plane and through its centre. Angular frequency of rotation is ω.

13. The tension in the ring will be
(A) $\frac{a\rho R^2\omega^2}{2}$
(B) $a\rho R^2\omega^2$
(C) $2a\rho R^2\omega^2$
(D) $\frac{a\rho R^2\omega^2}{4}$.

14. The ratio of kinetic energy to potential energy is
(A) $\frac{y}{\rho R^2\omega^2}$
(B) $\frac{2y}{\rho R^2\omega^2}$
(C) $\frac{y}{2\rho R^2\omega^2}$
(D) $\frac{y}{4\rho R^2\omega^2}$.

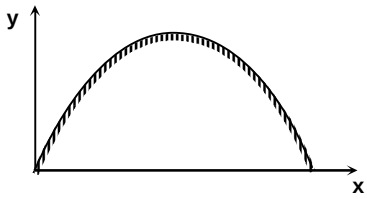
Space for Rough work

Multiple Correct Answer(s) Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

15. A reflecting surface is represented by the equation

$$y = \frac{2L}{\pi}\sin\left(\frac{\pi x}{L}\right),\ 0 \le x \le L$$

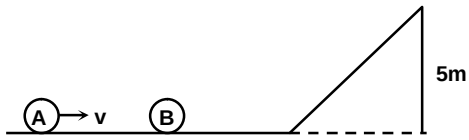


A ray travelling horizontally becomes vertical after reflection. The co-ordinate of the points where this ray is incident is

(A) $\left(\frac{L}{4}, \frac{\sqrt{2}L}{\pi}\right)$ (B) $\left(\frac{L}{3}, \frac{\sqrt{3}L}{\pi}\right)$

(C) $\left(\frac{3L}{4}, \frac{\sqrt{2}L}{\pi}\right)$ (D) $\left(\frac{2L}{3}, \frac{\sqrt{3}L}{\pi}\right)$

16. Two balls A and B of masses m and 2m respectively are lying on a smooth surface as shown in the figure. Ball A hits the ball B (which is at rest) with a velocity v = 16 m/s. Such that B just reaches the highest point of inclined plane. ($g = 10\ m/s^2$)

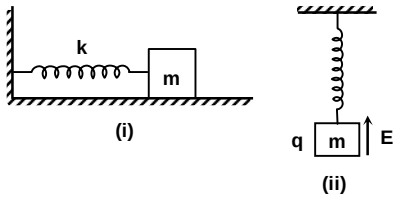


(A) e = 3/4 (B) e = 7/8

(C) $v_A = -4$ m/s (D) $v_A = 8$ m/s

where e is the coefficient of restitution and v_A is velocity of the ball A after collision.

17. A particle of mass m is executing SHM as shown in diagram (1) and (2). In diagram (2) particle contains charge q such that Eq = mg. If their velocities are same at mean position and let A_1 and A_2 be their amplitudes and T_1 and T_2 be their time periods respectively then



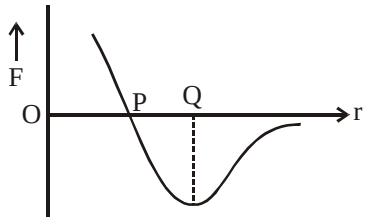
(A) $A_1 = A_2$ (B) $A_1 < A_2$

(C) $T_1 = T_2$ (D) $T_1 > T_2$

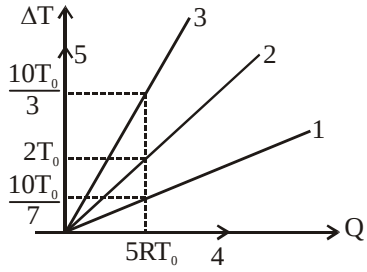
Space for Rough work

18. Two objects of mass m and 4 m are at rest at an infinite separation. They move towards each other under mutual attraction. If G is the universal gravitational constant. Then at separation r.
(A) The sum of energy of the two objects is negative.
(B) Their relative velocity of approach is $\left(\frac{10Gm}{r}\right)^{1/2}$ in magnitude
(C) The sum of kinetic energy of the objects is $\frac{4Gm^2}{r}$.
(D) The sum of angular momenta of both the objects is zero about any point.

19. Figure shows roughly how the force F between two adjacent atoms in a solid varies with separation 'r'
(A) OQ is equilibrium separation
(B) Hooke's law is obeyed around P
(C) The potential energy of the atom is the gradient of the graph at all points
(D) The energy to separate the atoms completely is obtained from the area enclosed below the axis or r.



20. Figure (Five straight lines numbered 1, 2, 3, 4 and 5) shows graph of change in temperature ΔT versus heat supplied Q for different processes performed on a gas of one mole.
(A) line 4 and 5 (coinciding with Q and ΔT axis respectively) represent isothermal and adiabatic processes respectively
(B) line 2 represents isochoric process for diatomic gas
(C) line 1 and 3 represent isobaric process for a diatomic gas and isochoric process for a monoatomic gas respectively
(D) line 2 represents isobaric process for a monoatomic gas.



Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

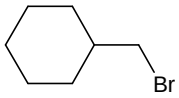
This section contains 8 multiple choice questions numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

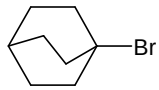
1. Which among the following is the strongest oxidizing agent?

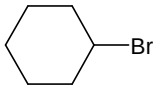
   (A) IO_4^- (B) XeO_6^{4-}

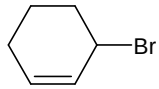
   (C) PO_4^{3-} (D) SO_4^{2-}

2. The compound which undergoes S_N1 reaction most rapidly is:

   (A) 

   (B) 

   (C) 

   (D) 

3. At 20°C, the osmotic pressure of a solution of 100 ml of which contains 6.33 g of the colouring substance of the blood haematin is 243.4 kP_a. What is the molecular formula of the haematin; if the elementary composition of haematin is C = 64.4%, H = 5.2%, N = 3.8%, O = 12.6% and Fe 8.8%?

   (A) $C_{34}H_{33}N_4O_5Fe$ (B) $C_{33}H_{32}N_4O_5Fe$

   (C) $C_{34}H_{30}N_3O_5Fe$ (D) $C_{32}H_{28}N_4O_5Fe$

4. Calculate E^o for the reactions:

   $ZnY^{2-} + 2e^- \rightleftharpoons Zn(s) + Y^{4-}$, where Y^{4-} is the completely deprotonated anion of EDTA, the formation constant for ZnY^{2-} is 3.2×10^{16} and E^o for $Zn \longrightarrow Zn^{2+} + 2e^-$ is 0.76V.

   (A) - 1.25 V (B) 0.48 V

   (C) + 0. 68 V (D) - 0.27 V

5. The freezing point of 0.20 M solution of weak acid HA is 272.5 K. The molality of the solution is 0.263 mol kg^{-1}. Find the pH of the solution on adding 0.25 M sodium acetate solution:

   (K_f of water = 1.86 K $kgmol^{-1}$).

   (A) 5.4 (B) 3.2

   (C) 6.1 (D) 4.11

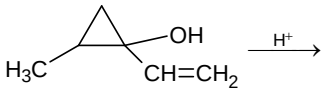
6. Aqueous solution of a (Y) salt is alkaline to litmus. On strong heating, it smells up to give a glassy material. When conc. H_2SO_4 is added to hot concentrated solution of (Y), white crystal of a weak acid separates out. Hence, the compound (Y) is:

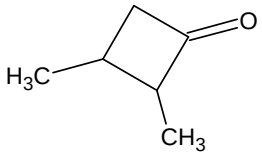
   (A) $Na_2SO_4.10H_2O$ (B) $Ca_2P_6O_{11}.10H_2O$

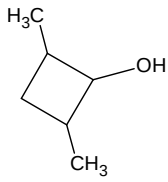
   (C) $Na_2B_6O_{11}$ (D) $Na_2B_4O_7.10H_2O$

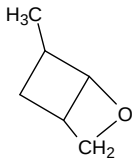
Rough Work

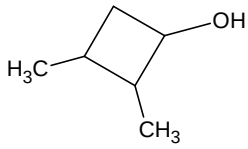
7. The major product of the following reaction is:

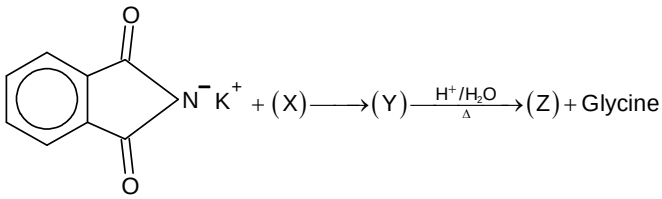


(A) 

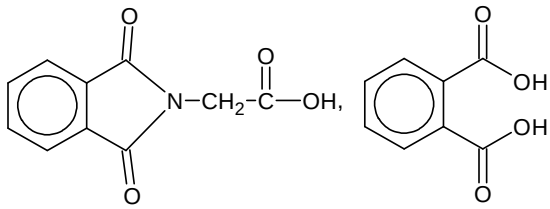
(B) 

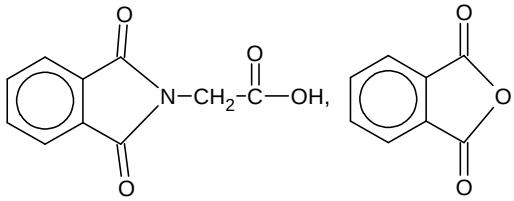
(C) 

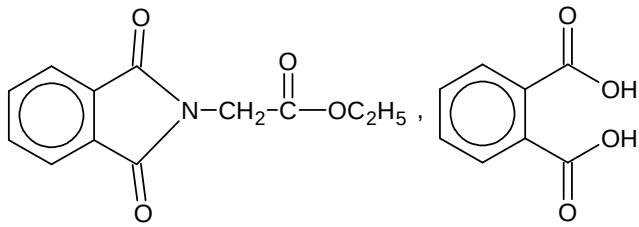
(D) 

8. 

Compound X, Y, Z are:

(A) Chloroacetic acid, 

(B) Bromo acetic acid, 

(C) Ethyl chloro acetic acid, 

(D) None of the above

Rough Work

Comprehension Type

This section contains 3 groups of questions. Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 9 to 10

Elements (A) burns in nitrogen atmosphere to give an ionic compound (B). Compound (B) gets hydrolysed by water giving (C) and (D). A solution of (C) becomes milky on passing carbon dioxide through it. Element (A) reacts quite readily with cold water liberating hydrogen gas and forming metal hydroxide.

9. Select the correct statement with respect to (D).
   (A) It is trigonal pyramidal in shape.
   (B) It can be obtained by the hydrolysis of calcium cyanamide, $CaCN_2$.
   (C) It gives precipitate with $HgCl_2$.
   (D) All of these

10. The compound (C) is:
   (A) $Mg(OH)_2$ (B) $Ba(OH)_2$
   (C) $Ca(OH)_2$ (D) Both (B) and (C)

Paragraph for Question Nos. 11 to 12

1 g of a weak monobasic acid (MW = 100) when dissolved in 100 g of liquid A (MW = 100) increases the boiling point from 80° C to 81° C at 760 mm of Hg. The vapour pressure of liquid A, at 90° C is 850 mm of Hg. Degree of dissociation of the monobasic acid is 0.189 at 90° C. Enthalpy of neutralization of a strong acid with strong base = - 57.3 kJ/equivalent.

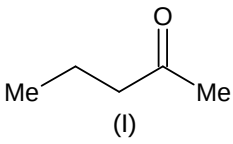
11. ΔH_v for the liquid A is (assuming it remain constant within the temperature range considered)
   (A) 11.99 kJ mol^{-1} (B) - 11.99 kJ mol^{-1}
   (C) 119.9 kJ mol^{-1} (D) - 119.9 kJ mol^{-1}

12. Degree of dissociation of the monobasic acid at 80^oC will be:
   (A) 0.2 (B) 0.158
   (C) 0.1158 (D) 0.52

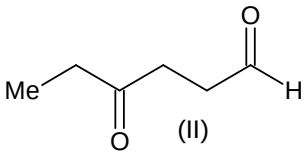
Rough Work

Paragraph for Question Nos. 13 to 14

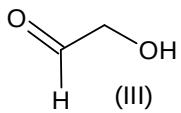
Compound (A) $C_{13}H_{24}O$ is a pheromone of codling moth. On hydrogenation, compound (A) absorbs two molar equivalents of H_2 and is converted to 3-ethyl-7-methyl decan-1-ol. On reductive ozonolysis, it gives the following compounds:



(I)



(II)



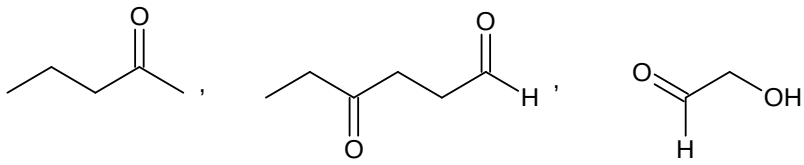
(III)

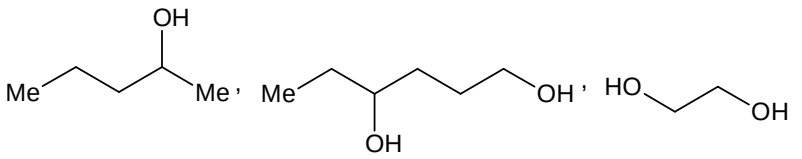
13. The number of stereoisomers of (A) is:

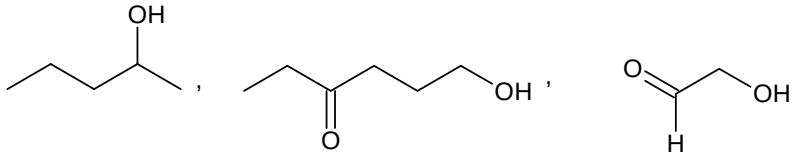
(A) 2 (B) 4

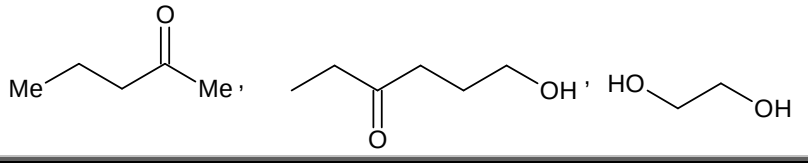
(C) 6 (D) 8

14. The products of (A) on ozonolysis followed by reduction with LAH are:

(A) 

(B) 

(C) 

(D) 

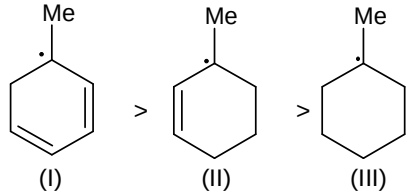
Rough Work

Multiple Correct Choice Type

This section contains 6 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

15. Which statements are correct about Lactose?
    (A) (C_1 - OH) of β-glucose is linked with (C_4) of galactose.
    (B) (C_1 – OH) of β-galactose is linked with (C_4) of β-glucose.
    (C) It exhibits mutarotation.
    (D) It is a reducing sugar.

16. Which of the following are correct?
    (A) Dipole moment of $NH_3 > NF_3$
    (B) Stability of free radical:



    (C) Basic strength : $CH_3O^- > OH^- > RS^-$
    (D) Basic strength : $I^- > Br^- > Cl^- > F^-$

17. A solution of colourless salt (X) on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt (s) X is (are):
    (A) NH_4NO_3 (B) NH_4NO_2
    (C) NH_4Cl (D) $(NH_4)_2SO_4$

18. If the rms velocity of nitrogen and oxygen molecule are same at two different temeperature and same pressure; then:
    (A) Average speed of molecules is also same
    (B) Density (gm/l) of nitrogen and oxygen is also equal
    (C) Number of moles of each gas is also equal
    (D) Most probable velocity of molecules is also equal

19. Which of the following statement(s) is/are true?
    (A) In metal carbonyl complexes, d_{C-O} increases compared to that in CO molecule
    (B) The pair of compounds $[Cr(H_2O)_6]Cl_3$ and $[CrCl_3(H_2O)_3].3H_2O$ show hydrate isomerism
    (C) d_{z^2} orbital of central metal atom/ion is used in dsp^2 hybridization
    (D) Facial and meridional isomers associated with $[Ma_3b_3]^{n\pm}$ type complex compound, both are optically inactive

20. A 0.1 M sodium acetate solution was prepared having $K_h = 5.6 \times 10^{-10}$.
    (A) The degree of hydrolysis is 7.48×10^{-5} (B) The OH^- concentration is 7.48×10^{-3} M
    (C) The OH^- concentration is 7.48×10^{-6} M (D) The pH is approximately is 8.88

Rough Work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If $6x^5 + 5x^3 + x^2 + 4x - 1 = a(x-1)^5 + b(x-1)^4 + c(x-1)^3 + d(x-1)^2 + e(x-1) + f$ then what is the value of $a + 2b + 3c + 4d + 5e + 5f$?

   (A) 780 (B) 820

   (C) 870 (D) 910

2. If $f(x) = \prod_{n=1}^{50} (x-n)^{n(51-n)}$; then $\frac{f(51)}{f'(51)}$

   (A) 4040 (B) $\frac{1}{1275}$

   (C) $\frac{1}{4040}$ (D) 1275

3. In a triangle ABC, area of triangle (Δ) and A are fixed. The angles B and C and length of sides a, b, c are variable quantities. If length of the side 'a' is minimum then

   (A) $c = \frac{\sqrt{2\Delta}}{\sin A}$ (B) $b = \sqrt{\frac{2\Delta}{\sin A}}$

   (C) $a = \frac{\sqrt{2\Delta}}{\sin A}$ (D) triangle is equilateral

4. If f(x) is differentiable function with $f'(0) = 1$ and
   $\cos 2x \sin 2y(f(3x + 3y) + f(3x - 3y)) = \sin 2x \cos 2y(f(3x + 3y) - f(3x - 3y))$ for all $x, b \in R$, then the equation of tangent to $y = f(x)$ at $x = \frac{\pi}{2}$ is

   (A) $2x + 4y = 3\sqrt{3} - \pi$ (B) $2x - 4y = \pi - 3\sqrt{3}$

   (C) $2x - y = 3\sqrt{3} - \pi$ (D) none of these

Space for rough work

5. Let $a_1, a_2, ... a_n$ be the terms of a G.P. whose common ratio is r. Set S_k denotes the sum of first k terms of the G.P. then the value of $\sum_{1\le i<j\le m} a_i a_j$ in terms of S_{m-1} and S_m is

(A) $r\, S_{m-1}\, S_m$ (B) $\frac{r}{1+r} S_{m-1} S_m$

(C) $\frac{S_{m-1} S_m}{1+r}$ (D) $S_{m-1}\, S_m$

6. $\lim_{x\to 0} \frac{\sqrt[3]{1+3x} - 1 - x}{(1+x)^{101} - 1 - 101x}$ has the value equal to

(A) $-\frac{3}{5050}$ (B) $-\frac{1}{5050}$

(C) $\frac{1}{5050}$ (D) $\frac{1}{4950}$

7. Let f(x) is a quadratic function such that f(0) = 1 and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function. Then the value of f′(0) is

(A) 2 (B) 3

(C) 4 (D) 5

8. Curves $ax^2 + 2hxy + by^2 - 2gx - 2fy + c = 0$ and $a'x^2 - 2hxy + (a' + a - b)y^2 - 2g'x - 2f'y + c = 0$ intersect at four concyclic points A, B, C and D. If P is the point $\left(\frac{g'+g}{a'+a}, \frac{f'+f}{a'+a}\right)$, the value of $\frac{PA^2 + PB^2 + PC^2}{PD^2}$ is

(A) 1 (B) 2

(C) 3 (D) 4

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 9 to 10

Read the following write up carefully and answer the following questions:

While finding the sine of a contain angle x, an absent minded professor failed to notice that his calculator was not in the correct angular mode. However he was lucky to get right answer. The two least positive value of x for which the sine of x degree is the same as the sine of x radian found by him as $\frac{m\pi}{n-\pi}$ and $\frac{p\pi}{q+\pi}$, where m, n, p and q are positive integers. Suppose $\frac{mn}{pq}$ be denoted by quantity 'L'

9. The value of (m + n + p + q) is equal to
   (A) 720 (B) 900
   (C) 1080 (D) 1260

10. If x is measured in radians and $\lim_{x\to\infty}\left(\sqrt{Ax^2+Bx}-Cx\right)=2$, the value of $\frac{BC}{A}$ equals (A, B, C $\in$ R)
   (A) 4 (B) 2
   (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:

Square ABCD has vertices A(1, 4), B(5, 4), C(5, 8) and D(1, 8) from a point P outside the square, a vertex of square is said to be visible if it can be connected to P by a straight line that does not pass through square. Thus from any point P, outside either two or three of the vertices of the square are visible. The visible area of P is the area of one triangle or the sum of the area of the two triangle formed by joining P to the two or three visible vertices of the square

11. Visible area of the point P with coordinate (2, –6) in square unit is
   (A) 10 (B) 20
   (C) –30 (D) 40

12. Visible area of the point Q(11, 0) in square unit is
   (A) 20 (B) 30
   (C) 40 (D) 50

Space for rough work

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

Let $f(x)=\begin{cases} e^{\{x^2\}}-1 & x>0 \\ \dfrac{\sin x-\tan x+\cos x-1}{2x^2-\ln(2+x)+\tan x} & x<0 \\ 0 & x=0 \end{cases}$ (where {} represents fractional part function). Lines L_1 and L_2 represent tangent and normal to curve y = f(x) at x = 0. Consider the family of circles touching both the lines L_1 and L_2

13. Ratio of radii of two circles belonging to this family cutting each other orthogonally is
(A) $2+\sqrt{3}$ (B) $\sqrt{3}$
(C) $2+\sqrt{2}$ (D) $2-\sqrt{2}$

14. A circle having radius unity is inscribed in the triangle formed by L_1 and L_2 and a tangent to it. Then the minimum area of the triangle is
(A) $3+\sqrt{2}$ (B) $3-\sqrt{2}$
(C) $3+2\sqrt{2}$ (D) $3-2\sqrt{2}$

Multiple Correct Choice Type

This section contains **6 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

15. Consider the points $A(\vec{o})$, $B(3\hat{i}+f(t)\hat{j}+\hat{k})$ and $C(2\hat{i}+f'(t)\hat{j}+2\hat{k})$ and another point D are coplanar for all $t \in R$
(A) if D is $(\hat{i})$ then $f(x) = ce^{2x}$, $c \in R$ (B) if D is $(\hat{i})$ then $f(x) = c\ln(2x)$, $c \in R$
(C) if D is $(\hat{i}+\hat{j})$ then $f(x) = 2 + ce^{2x}$, $x \in R$ (D) if D is $(\hat{i}+\hat{j})$ then $f(x) = 2 + c\ln(2x)$, $c \in R$

16. Let $u_n=\int_0^{\pi/2}\cos^n x\cos nx\,dx$ it turns out that u_1, u_2, u_3, ... form a G.P. then
(A) the common ratio of G.P. is $\frac{1}{2}$ (B) the common ratio of G.P. is $\frac{1}{4}$
(C) then $u_n=\frac{\pi}{2^{2n}}$ (D) then $u_n=\frac{\pi}{2^{n+1}}$

Space for rough work

17. Consider the following three planes
$P_1 : x = y \sin\alpha + z \sin\beta$
$P_2 : y = z \sin\theta + x \sin\alpha$
$P_3 : z = x \sin\beta + y \sin\theta$
If P_1, P_2, P_3 intersect in the line $\frac{x}{\cos\theta} = \frac{y}{\cos\beta} = \frac{z}{\cos\alpha}$ then the value of $\alpha + \beta + \theta$ is equal to
(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{2}$
(C) $-\frac{3\pi}{2}$ (D) $\frac{7\pi}{2}$

18. Five points are selected on a circle of radius r, consider the centres of the five rectangular hyperbola each passing through four of these five points, then
(A) these centres lie on another hyperbola
(B) these centres lie on a circle of radius $\frac{r}{2}$
(C) these centres lie on a circle of radius 2r
(D) mean of these four points is mid–point of centres of circle and rectangular hyperbola

19. Which of the following condition must be satisfied so that two of the lines represented by the equation $ay^4 + by^3x + cy^2x^2 + dyx^3 + ax^4 = 0$ will bisect the angle between the other two?
(A) b + d = 0 (B) bd = 1
(C) c + 4a = 0 (D) c + 6a = 0

20. Let $P_k (k = 1, 2, ... n)$ be the nth root of unity. Let z = a + ib and $A_k = Re(z)\, Re(P_k) + i\{Im(z)\, Im(P_k)\}$ then which of the following is true
(A) A_k lies on ellipse
(B) A_k lies on hyperbola
(C) if S be the focus of locus of A_k then $\sum_{k=1}^{n} A_k S = na$
(D) if S be the focus of locus of A_k then $\sum_{k=1}^{n} (A_k S)^2 = \frac{n}{2}(3a^2 - b^2)$

Space for rough work